

CAPE LOOKOUT, NC, USA

WMO: 994160

Lat: 34.622N

Lon: 76.525W

Elev: 15

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.7	31.8	9.7	9.1	32.3	14.0	11.2	35.7	34.2	49.1	32.1	46.0	17.9	350	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	85.2	78.7	84.2	78.2	83.3	77.7	81.0	83.4	80.1	82.8	79.3	82.3	12.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	157.5	82.5	79.3	152.0	81.9	78.3	147.0	81.6	44.5	83.3	43.5	82.7	42.7	82.3	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.9	29.9	26.1	21.0	88.1	4.9	1.5	17.5	89.2	14.7	90.0	11.9	90.8	8.4	91.9		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.4	47.4	48.5	54.3	62.3	69.5	76.7	79.8	79.3	75.8	68.0	57.9	52.1
	DBStd	13.08	9.02	7.44	7.05	5.65	4.75	3.81	2.33	2.65	3.52	6.19	7.46	8.04
	HDD50	396	157	105	40	2	0	0	0	0	0	0	19	73
	HDD65	2151	546	461	335	113	18	1	0	0	43	230	404	
	CDD50	5649	76	64	173	370	606	802	924	907	775	557	257	138
	CDD65	1928	0	0	2	31	159	353	459	442	325	135	18	4
	CDH74	13894	0	0	0	7	254	2453	4472	4213	2110	381	4	0
	CDH80	1869	0	0	0	0	7	240	759	711	148	4	0	0
Wind	WSAvg	13.4	13.9	14.7	15.5	14.9	13.3	11.8	11.6	11.8	13.6	12.7	13.3	13.3
Precipitation	PrecAvg	56.0	4.3	3.7	3.8	3.3	3.9	4.1	6.2	7.4	7.5	4.3	3.8	4.3
	PrecMax	84.5	8.5	11.0	11.6	8.9	13.0	9.1	13.5	15.0	15.8	11.9	9.7	7.6
	PrecMin	35.7	1.7	1.3	1.3	0.7	1.0	0.3	2.6	2.3	1.5	0.1	1.1	1.8
	PrecStd	10.9	1.9	2.1	2.0	2.1	2.5	2.0	2.5	3.3	4.1	2.8	2.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.5	66.5	70.2	75.2	80.9	85.8	86.4	86.5	84.0	80.4	74.4	71.3
		MCWB	64.3	63.9	64.6	69.6	75.3	77.4	79.6	79.1	76.9	75.6	70.9	69.3
	2%	DB	65.1	63.7	67.7	72.8	77.9	83.9	85.1	85.2	82.6	79.0	72.5	68.5
		MCWB	62.5	60.9	63.8	68.3	73.4	77.3	79.2	78.7	76.5	75.2	69.4	66.4
	5%	DB	62.7	61.3	65.9	71.1	76.7	82.4	84.2	84.1	81.7	77.8	70.7	66.2
		MCWB	60.2	59.0	62.5	67.1	72.4	77.0	78.6	78.1	76.1	73.8	67.3	63.9
	10%	DB	60.2	59.0	64.1	69.3	75.3	81.4	83.3	83.2	80.8	76.5	68.8	64.1
		MCWB	57.7	56.2	60.9	65.5	71.3	76.7	78.1	77.5	75.5	72.1	65.3	62.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.1	64.4	67.0	71.3	77.1	81.1	81.6	82.3	81.5	79.1	72.1	70.4
		MCDB	67.4	65.6	68.3	73.2	79.1	82.6	84.0	85.2	81.5	79.3	73.8	71.5
	2%	WB	63.7	62.2	65.3	69.8	75.3	79.7	80.6	81.3	79.8	77.0	70.5	68.1
		MCDB	65.3	63.0	67.1	72.3	77.5	82.2	83.4	84.2	80.6	78.0	72.3	69.3
	5%	WB	61.5	59.6	63.7	68.5	74.0	78.5	80.0	79.9	78.3	75.2	68.4	65.3
		MCDB	63.4	60.8	66.0	70.7	76.3	81.2	83.0	83.4	80.3	77.3	70.1	66.4
	10%	WB	59.0	57.0	61.7	67.0	72.8	77.5	79.3	78.7	77.3	73.4	65.9	63.0
		MCDB	61.1	59.0	64.0	69.0	75.1	80.3	82.6	82.7	80.0	76.0	68.1	64.5
Mean Daily Temperature Range	5% DB	MDBR	11.3	10.6	9.7	8.0	6.5	5.5	5.6	5.7	5.9	7.8	9.8	10.5
		MCDBR	12.3	11.7	9.9	8.0	6.6	6.2	6.0	6.0	5.9	6.7	9.9	11.0
		MCWBR	10.7	12.0	7.9	6.0	4.4	4.3	4.5	4.2	5.2	5.8	9.7	10.6
	5% WB	MCDBR	11.1	11.4	9.7	7.9	6.1	5.5	5.8	6.1	5.7	6.1	9.6	10.5
		MCWBR	10.7	12.5	9.0	7.2	5.0	4.5	5.0	4.9	5.4	6.3	10.5	10.8
Clear-Sky Solar Irradiance	taub	0.330	0.339	0.363	0.401	0.446	0.487	0.512	0.496	0.424	0.374	0.350	0.340	
	taud	2.538	2.511	2.460	2.353	2.273	2.203	2.144	2.185	2.391	2.504	2.519	2.532	
	Ebn at Noon	279	288	289	281	268	256	249	251	266	272	268	267	
	Edh at Noon	26	30	34	39	43	46	49	46	36	30	26	25	
All-Sky Solar Radiation	RadAvg	808	1045	1389	1764	1962	2024	1905	1737	1432	1219	901	721	
	RadStd	48	100	105	127	129	114	120	102	94	117	63	52	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (27 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+211	

Nomenclature: See separate page